

Overview of Architectures for Satellite's Regulated Bus Power System

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Abstract -- The power supply subsystem of satellite is the core constant, throughout the whole flight stage. The solar array-battery power system is the main selection for the regulated bus power supply system architecture. The power conditioning unit is responsible for bus voltage regulating and energy management, thus supplying the high-performance steady bus to spacecraft payload. The architecture of the power subsystem must consider system overall efficiency, bus static output characteristics, reliability, modularity, power density, even control methods and other technical specifications. In this paper, the working principle and performance of different power supply system architectures are analyzed, concerning the solar array power regulation, the charging and discharging regulation of battery, the battery charging control strategy, and the bus voltage stabilization. Therefore, the power system designers can make choice of a power system architecture according to the different tracks, load characteristics, solar array configuration, battery configuration, satellite's weight, and thermal control system.

Index Terms-- Battery ; Power sub-system; Power conditioning; Regulated bus; Solar array

I. INTRODUCTION

AS the essential and even the only power supply unit in satellite, once the fault of the satellite power supply system occurs, the whole satellite will failure, and the end of the space mission directly. The weight of the entire power system accounts for about 28%~33% of the satellite, while the generator, energy storage device and power control unit (PCU) each account for about 33% of the entire spacecraft power system [1-5].

By the mode of bus voltage regulation, the solar array-battery supply control architecture is divided into three categories: unregulated bus, which is mainly used in China's early spacecraft and some small satellites at present; partially regulated bus whose voltage is unregulated during an eclipse, and is regulated by shunt control during sunlight, to maintain the stability of the bus; and regulated bus, which is always stable through internal control either in sunlight or in eclipse mode, and its voltage fluctuates little with the fluctuation of load power. The fully regulated bus can provide power for the load in high performance while achieving high modularization and a power extension. A regulated bus is currently adopted in most communications satellites, navigational satellites, and remote sensing satellites to control voltage and supply.

The fully regulated bus architecture is thus one of the studies focuses in space power system. Besides, the selection

of architecture is based on the analysis of key factors such as different orbital environment, energy storing device or generator controlled, space environment, weight, and so on, in a bid to ensure a more efficient, reliable and modularized, and higher power density and bus performance of the whole power system.

What's more, lasers and communications fired from the pulse power remote sensing-applied loads and high power and repetitive pulse loads also bring new problems to the power supply system in spacecraft: if simpler unregulated bus architecture and larger battery capacity are applied to power the platform and the payload, the platform is vulnerable to electromagnetic interference from the pulse power load, and the unadjusted bus will also complicate the design of the power subsystem, including the secondary power supply. For the regulated bus architecture such as S3R, the solar array and battery have to be sufficiently equipped meeting the peak power, which is extremely costly. At the same time, it is impossible for the subsystem to configure components in the same way [6].

Power system architectures can be classified as Peak Power Tracker (PPT) and Direct Energy Transfer (DET) by the energy transfer form of the power control unit to the solar array. PPT system is an enabling technology that can maximize spacecraft solar array output power at the beginning of its life, which is helpful to optimize the solar panel configuration and reduce the size and weight of the spacecraft [7-10]. Thales Alenia Space present implementing analog maximum power point tracking (MPPT) algorithm. The used analog MPPT algorithm does not need to compute the solar array power and is compatible with both regulated and unregulated bus [11-12].

This paper presents the application scenarios, advantages and disadvantages, and design considerations of different architectures. Meanwhile, based on the application characteristics of the above-mentioned fully regulated power system, the existing satellite power control unit architectures at home and abroad will be summarized and analyzed.

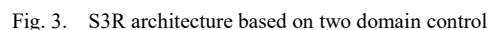
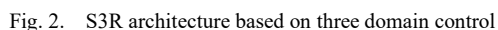
II. DET POWER CONTROL ARCHITECTURE

A. S3R Architecture Family

Featuring simplicity, reliability and high efficiency, Sequential Shunt Switching Regulator (S3R) architecture is widely used in the design of power control [13]. For example, in Thales Alenia Space Belgium (ETCA), its first- and second-generation products both apply S3R architecture.

However, when applying S3R architecture, the adoption of solar array parasitic capacitance has to be carefully analyzed.

S3R architecture is generally based on three do-main control, as shown in Fig.2. The lower the main error amplifier (MEA) voltage, the higher the S3R and battery discharge regulator (BDR) output power and the much lower the battery charge regulator (BCR) output power. MEA may regulate the bus stability according to different load, generation power and other factors in different domains.



The high voltage and high power (21kW) power control unit based on S3R adopts a novel shunt topology and control method to solve the problems of larger parasitic capacitance and its related negative effects caused by the solar array made of new materials [21].

At the architecture level, the solar array can also be divided into two parts: low-current solar array and high-current solar array. How the architecture operates is as follows: a low-current solar array is always in the state of switch regulating; and a high-current solar array is in the power or shunt state, but will not work in the state of switch regulating. Because the only low-current solar array is in the state of switch regulating, the MOSFET power loss in the switching circuit will be reduced. In addition, the lower solar array current means much lower parasitic capacitance, thus the bus steady-state characteristics and EMC problems will be improved, weakening the impact of triple-junction gallium arsenide solar battery and large single-array current on S3R. Parallel power process S3R architecture is shown in Fig.4 [22], [23].

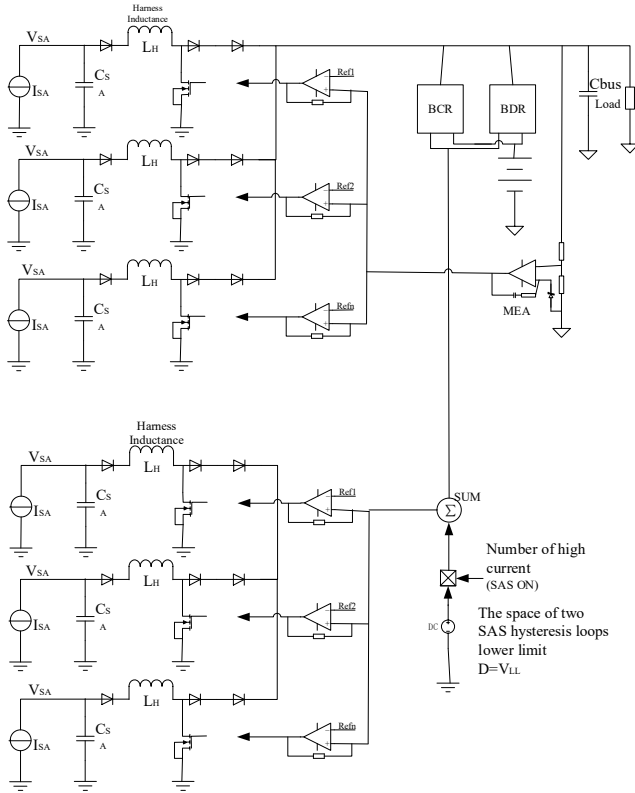


Fig. 4. Parallel power process S3R architecture

The disadvantages of the architecture: the current design of the solar battery needs to be changed, and the SADA system needs to be adjusted accordingly. Also, there are too many interfaces between PCU and solar cells.

To improve the power density of the system, a Battery Charge Discharge Regulator (BCDR) module is used to replace the original BCR and BDR. BCDR replaces diodes in the traditional buck-boost topology with MOSFET devices to enable bidirectional current. Also, the protective circuits, part of control circuits, auxiliary sources and so on can be shared, so as to improve the power density. In addition, the integration of BCDR can effectively avoid the circulation problems of BCR and BDR in traditional S3R. It is necessary to set the number of BCDR based on the maximum load power during the eclipse and taking considering of the reliability in PCU. In most of medium and high orbits, the charging power is much less than the discharge power, and the peak at charging power of a single BCDR can be the same as the discharge power, so the failure of one BCDR will only reduce the discharge power but will not affect the total charging power. Therefore, it will not cause a significant reduction in reliability. S3R architecture merging BCR and BDR is shown in Fig.5 [24]-[26].

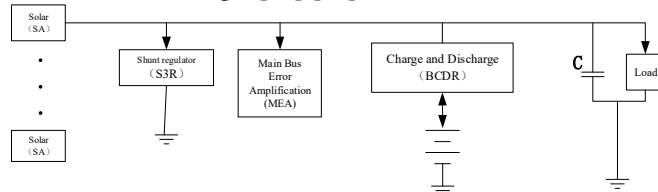


Fig. 5. S3R architecture merging BCR and BDR

It is feasible for BCDR to replace the original BCR and

BDR for traditional S3R architecture based on three domain control. It's worth mentioning that, in some cases where the traditional S3R architecture is based on two domain control (the top of the BCR domain and the BDR domain overlaps), battery charge management (BCM) controls BCR and BDR to regulate the bus voltage and the charging power concurrently. It is obvious that the S3R architecture merging BCR and BDR cannot function in two domain control but three due to the unified topology, however, it can avoid the existence of circulation and reduce the power consumption of the system.

B. S4R Architecture Family

In the S3R architecture, an independent BCR function. After orbit determination and energy balance calculation, if the charging power increases, S3R can only solve this problem by increasing the number of BCRs, which will inevitably lead to a further reduction of the power density of the system. S4R charges the battery with an in series the power control unit, replacing the BCR in S3R, which could improve con-version efficiency and reduce heat consumption. Finally, it is realized that the weight of the solar array will not increase linearly due to the increase of charging power and excessive BCRs. Moreover, the number of inductors is much less compared with S3R, thus improving power density and efficiency. The design of the ordered switch also improves the reliability of the system, and its redundancy is beneath the threshold that the failure of a single unit will not impact the whole system. The S4R architecture is shown in Fig.6.

The principle of the S4R architecture charge control determines that the charging current of the battery as the pulse current, and the pulse amplitude is the single solar array current, of which the frequency and duty cycle are determined by factors such as control loop parameters and the single solar array current. Therefore, it is not suitable for scenarios with low requirements for battery charging current ripple. [16], [19], [27].

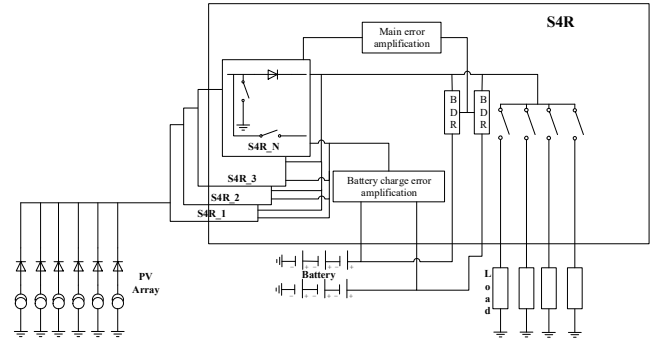


Fig. 6. S4R architecture power system

S4R architecture is generally based on two domain control, as shown in Fig.7. The lower the MEA voltage, the larger the S3R and BDR output power and the much less solar array connected to the battery. The solar array that supplies the bus is given priority, so there is no region where the charging power can be used to regulate the bus. MEA may regulate the bus stability according to different loads, generation power and other factors in different domains.

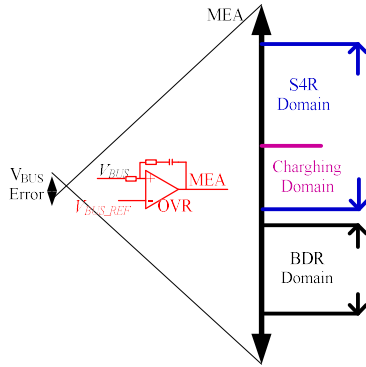


Fig. 7. S4R architecture based on two domain control

In addition, the fully regulated intermediate bus architecture can also be derived on the basis of unregulated bus architecture. The former architecture manages battery charge through the first stage S3R, and generates a fully regulated bus through the later stage DCDC, as shown in Fig.8.

This architecture can enable the battery bus to bear pulse power load by increasing battery capacity. However, the wide range of voltage change and the clamping solar battery voltage by the battery voltage will lead to a lower use ratio of solar battery, plus the later stage DCDC being in the working state, so the later stage DCDC needs to be more reliable.

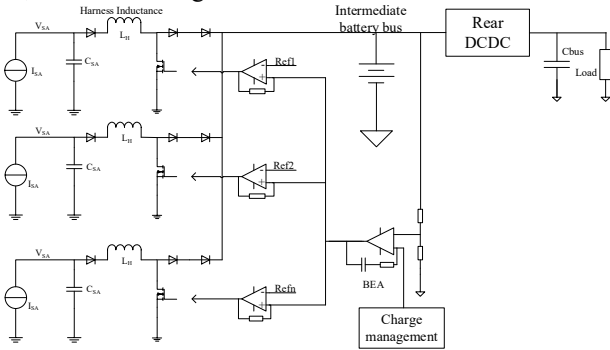


Fig. 8. Fully regulated architecture with central battery bus

The Diversion architecture developed by Astrium has been further improved on the basis of S4R architecture, which not only removes BCR but also avoids the use of battery charging power control in S4R, thus the power density can be further improved. Diversion architecture is shown in Fig.9.

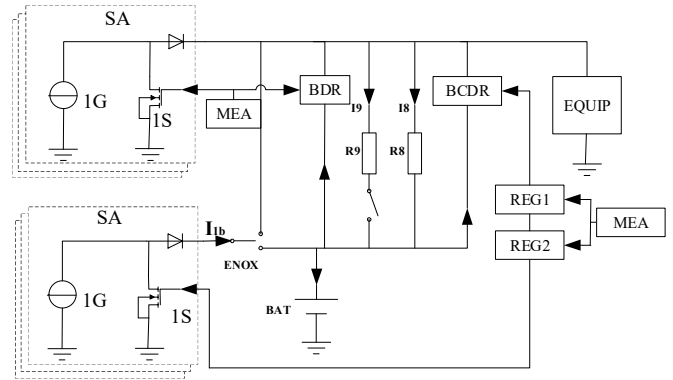
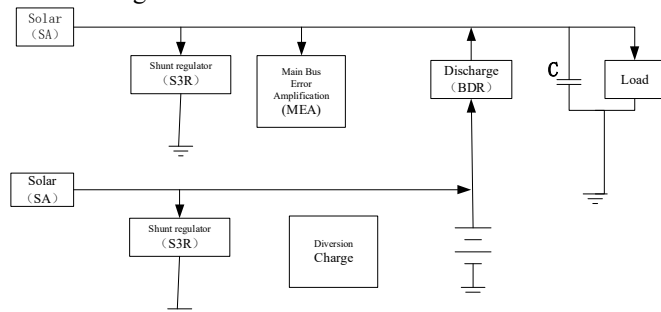


Fig. 9. Power system based on Diversion architecture

The basic principle of Diversion architecture is that charging array and BDR input current are subtracted to control the charging current. It controls the generation power by the charging array shunt, and regulates and controls charging current by BDR input current (through the Diversion charging control unit). In a discharge region, all BDRs obtain power from the battery and provide bus current together, so the BCR part of S3R can be completely removed through charging current control in the Diversion architecture. In Fig.9, BCDR and BDR have essentially discharge regulators, with the same internal module, but controlled by different control signals: BDR is directly controlled by MEA and only for discharge, while BCDR, controlled by MEA and regulator 1, whose power only flows from battery to bus. On this basis, both the discharge regulate bus and regulate charging current functions can be taken into account [28],[29].

The disadvantages of Diversion architecture: the storage battery charging is controlled by discharge current control, which complicates the control logic and bus stabilization; the charging array voltage is clamped by the battery voltage, resulting in the lower power use ration when the battery voltage is low; the energy of the charging array can only be supplied to the bus through BDR, and the efficiency is lower than the traditional way of transferring through DET. The equinox ENOX switch and supplementary charging resistors have excessive weight and will lead to power dissipation (after the ENOX switched to the bus, the battery self-discharge power needs to be replenished by R8 and R9).

Therefore, the Diversion architecture can be further improved (Fig.10): the S3R shunt shall be changed to S4R, thus the power can be transferred directly to the bus through the bus power supply branch. This way could raise efficiency and also remove the heavier equinox ENOX switch and supplementary charging resistors R8 and R9. The charging array S4R circuit preference to be controlled by MEA, followed by the Diversion regulator which controls BDR to regulate the bus voltage stabilization and discharge [30].

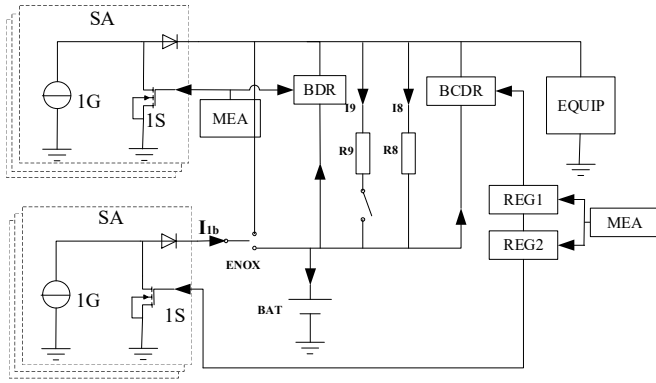


Fig. 10. Power system based on improved Diversion.

III. POWER SYSTEM BASED ON MPPT ARCHITECTURE

The most typical PPT system can control the output energy of solar array according to the power requirements of the spacecraft by connecting in series regulator between the solar array and the load, and effectively maximize the output power of the solar array during its life.

As shown in Fig.11, in the PPT regulated power control architecture, the SA terminal uses a boost, buck, or boost-buck power converter to replace the shunt in the S3R architecture, thus the MPPT can control the buck or buck converter PPT and realize peak power tracking or bus regulation [31]-[34].

The adoption of the SA terminal power converter in the architecture can eliminate the influence of input parasitic capacitance and realize MPPT at the same time. However, compared with the traditional S3R shunt, the power converter is much bigger and heavier, with lower power density than S3R architecture, and lower conversion efficiency than that of DET mode.

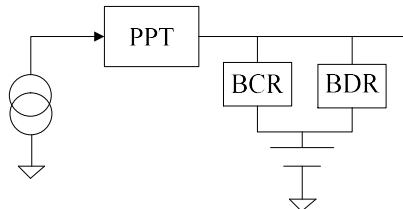


Fig. 11. Power system based on MPPT architecture.

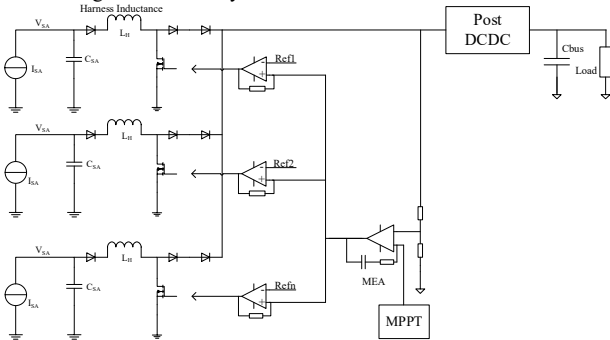


Fig. 12. Power system based on S3MPR.

In order to improve the efficiency of MPPT systems, the DET circuit is also used to realize MPPT in some systems without regulated architecture (Fig.12). Compared with the traditional S3R circuit, its bus voltage is no longer fixed, but controlled by the MPPT module that achieves peak power tracking by changing bus voltage. As a result, the output

voltage of the shunt varies widely, and the fully regulated bus is enabled through the later stage DCDC conversion [35]-[37].

The adoption of DET shunt circuit improves the performance of the architecture that needs a later stage DCDC to regulate voltage. In addition, the architecture is not suitable for scenarios with storage batteries, so it requires a stricter condition.

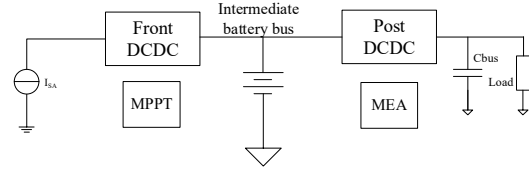


Fig. 13. MPPT architecture with central battery bus.

The fully regulated shunt architecture based on the central battery bus (Fig.13) can replace the front-stage shunt to a DCDC converter to withstand voltage difference between the solar battery and the storage battery to achieve MPPT. The improved architecture inherits the advantages of the pulse power load, and the power density is improved to some extent compared with that in Fig.11 [38].

In many cases, the power converter of space power supply can be abstracted as a problem of topology, control, energy management, and scheduling in multi-input (multi-source), multi-energy storage devices and multi-output power systems. Multi-port or three-port power converter is an effective method to solve this problem. Three-port power converter is capable of multi-input and multi-output single-stage power processing, which could improve the reliability, power density and conversion efficiency of a power converter, and avoid the direct energy transfer from solar battery to the bus, resulting in the MPPT control. Moreover, the parasitic capacitance of solar battery will be equivalent to the input filter capacitance of the converter, so the larger parasitic capacitance will not cause a negative impact on the power controller. The shunt module, battery charge regulator module and battery discharge regulator module in traditional S3R architecture are placed in a multi-port power topology to realize a power controller with integrated architecture. Compared with the traditional S3R architecture, its solar array, storage battery, and load are connected to the same topology with multi-port, which could extremely reduce the size and weight of the power controller and increases power density [39]-[41], as shown in Fig.14.

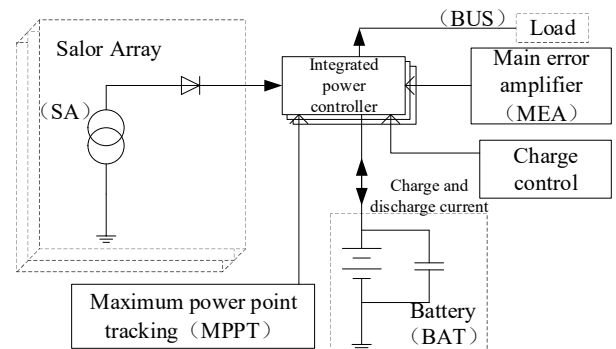


Fig. 14. United architecture power system based on three port converters.

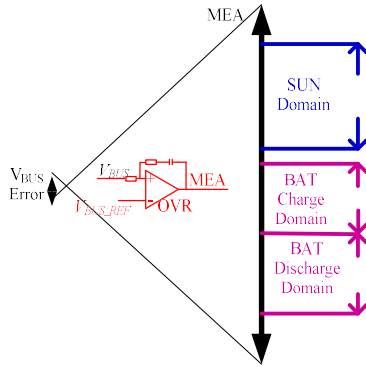


Fig. 15. Region distribution under MEA control.

Region distribution under MEA control is shown in Fig.15, in which the three-port converter is controlled to work in different modes for different MEA output voltage. Several modes can be automatically switched under the proper distribution of the control region.

IV. POWER CONTROL ARCHITECTURE ADOPTED TO PULSE POWER LOAD

In order to adapt to the second or minute pulse power load, it is vital that investigate the power control architecture adopted to pulse power load. The dual-bus (battery bus and platform bus) architecture shown in Fig.13 can adapt to the pulse power load by increasing the storage battery capacity, and the virtue of which is that the system uses only one set of storage batteries. However, this architecture requires the ongoing operation of a later stage DCDC, which requires high reliability. Meanwhile, the electromagnetic interference of the pulse power load in the front-stage battery on the later-stage platform bus should also be considered.

Fig.16 is another power control architecture adopted to pulse power load, which is commonly used basis on a fully regulated bus power supply system. It charges the secondary battery (storage battery with higher discharge ratio, ultracapacitor, or other energy storage devices) through the second-stage DCDC at the primary bus later stage, and the pulse load is directly hung on the secondary battery. The advantage of this architecture is that the pulse load is relatively far from the primary bus, and the system EMC is better. Its disadvantage is that two sets of energy storage devices are needed, which complicates the system. In addition, the pulse load bus is unregulated, which is not conducive to the design of pulse power load secondary power supply and its power distribution.

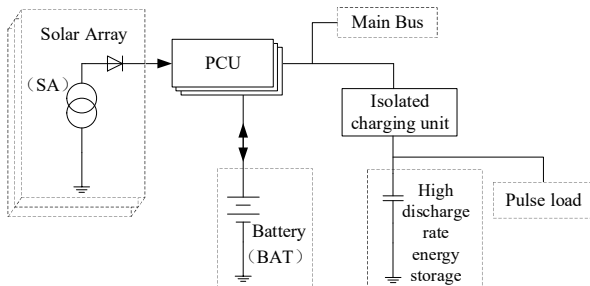


Fig. 16. Unregulated pulse power load architecture.

The architecture shown in Fig.17 can combine the

platform bus and the pulse power load bus into one, which requires another two units: the ultracapacitor charger CCR and the ultracapacitor discharger CDR. Therefore, the bus can be regulated and controlled to stabilize the voltage in the full operating range. This architecture can still adapt to the fully regulated architecture based on three domain control, adding CCR and CDR domains to realize bus voltage regulation and ultracapacitor charge and discharge management (as shown in Fig. 18). Different energy storage devices can be chosen to replace ultracapacitors in different scenarios [42].

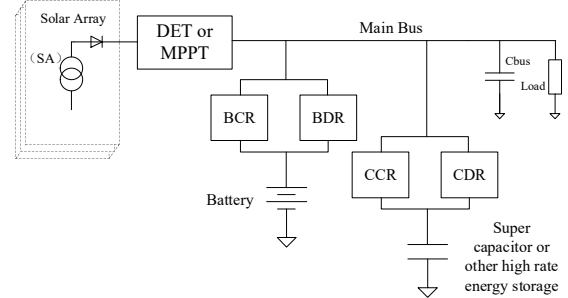


Fig. 17. Regulated mixed load architecture.

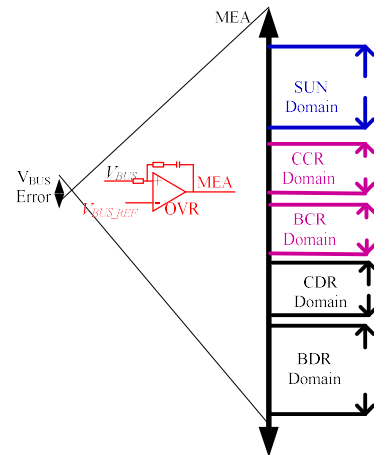


Fig. 18. S3R architecture based on three domain control.

The advantage of this architecture is that the pulse load bus voltage can also be fully regulated and reduce the design requirements of secondary power supply and power distribution. Besides, the use of high-performance energy storage devices such as ultracapacitor avoids large-capacity storage batteries and makes the system lighter and more cost-effective. But the platform and the pulse load sharing a common bus may lead to significant electromagnetic interference in case of steeper rising or falling edge of the pulse load. It is recommended to use the architecture in the pulse load with lower electromagnetic interference.

V. ARCHITECTURE WITH HIGH DYNAMIC PERFORMANCE

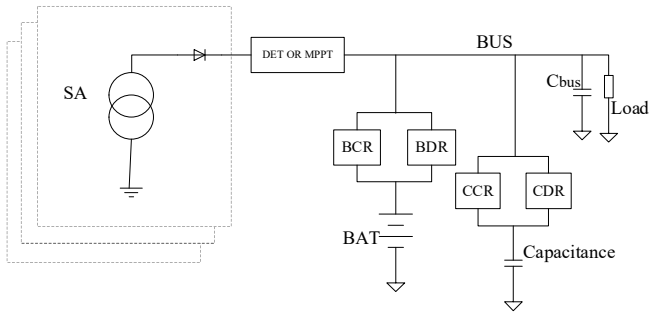


Fig. 19. PCU architecture with high dynamic performance.

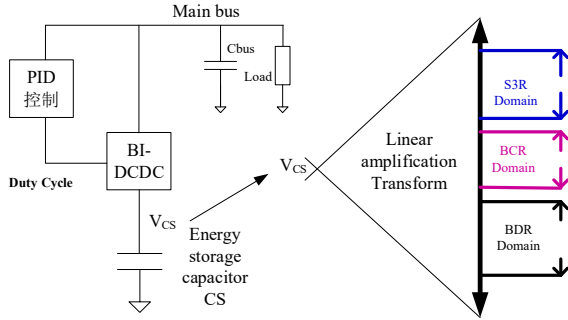


Fig. 20. Three domain control without MEA.

Load on TDMA (time division multiple access) is common in communication satellites. In order to improve the load adaptability of TDMA, the ultracapacitor in Fig.17 can be replaced with a conventional energy-storage capacitor to enhance the dynamic performance of loaded TDMA. However, the conventional five domain control based on MEA will not be able to meet the dynamic response requirements. A method with high dynamic response control without MEA is proposed in the document [39]. The VCS is used instead of MEA voltage as the control signal of DET (or MPPT) solar regulator, BCR and BDR, while the bi-directional DCDC is connected to the energy-storage capacitor CS directly regulates bus voltage. The system architecture and corresponding control mode as shown as Fig.19 and Fig.20. The power control output bus under this architecture and control method has good dynamic response performance and removes the MEA control unit. A bi-directional DCDC unit needs to be added to improve the dynamic response, and the energy-storage capacitor has to be self-healing to avoid a single point of failure [43].

VI. CONCLUSION

With the development of aerospace technology, requirements, such as power, dynamic response, pulse load and bus quality are increased and put forward for the power system. The power system architecture is also a vital part that has to be considered at the beginning of satellite design. This paper summarizes the architectures of the mainstream and the others, including the direct energy transfer, peak power tracker, power architecture adapted to pulse load, as well as high dynamic response power architecture. Meanwhile, their advantages, disadvantages, and application scenarios are also analyzed. It provides benefit reference and basis for the designers while configuring the satellite power system and component.

TABLE I
TYPICAL ARCHITECTURES CONTRAST

architecture	S ³ R	S ⁴ R	MPPT
Battery voltage	Unlimited	Below bus voltage	Unlimited
SA efficiency	High	High	The highest
Battery charge efficiency	High (BCR module)	The highest	High (BCR module)
Power loss	High	Low	The highest
Complexity	simple	complex	The most complex
Power density	Low	High	Low
Application	GEO satellites with low charging currents	GEO and LEO satellites with high charging currents	Environmental factors (The solar light intensity, operating temperature etc.) change SA U-I characteristic

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VII. BIOGRAPHIES

A technical biography for each author must be included. It should begin with the author's name (as it appears in the byline). Please do try to finish the two last columns on the last page at the same height. The following is an example of the text of a technical biography:

Hongyu Zhu was born in Harbin, China, in 1985. He received the B.S. degree from Northeast Forestry University, Harbin, Heilongjiang, in 2007, and the M.S. and Ph.D. degrees from Harbin Institute of Technology Shenzhen Graduate School, China, in 2009 and 2015, respectively. Since 2015, he has been a senior engineer at Shenzhen Aerospace New Power Technology Co., Ltd. His research interests include space power electronics, renewable energy systems and motion control. In these research fields, he was leading several industrial and government projects.

Bowen Zhang was born in China in 1988. He received the B.E. degree in electrical engineering from Huazhong University of Science and Technology, Wuhan, China, in 2010. He is currently an Engineer in power supply applications with the Shenzhen Academy of Aerospace Technology, Shenzhen, China. His research interests include power electronics modeling and control, renewable energy generation, and high-power-density converters.

Donglai Zhang (M'03) was born in Jilin, China, in 1973. He received the B.S., M.S., and Ph.D. degrees from Harbin Institute of Technology, Harbin, China, in 1994, 1996, and 1999, respectively. Since 2005, he has been a Professor with the Shenzhen Graduate School of Harbin Institute of Technology, Shenzhen, China. His research interests include analysis, modeling, and control of power converters, digital control techniques for power electronic circuits, and grid-connected converters for renewable energy systems. In these research fields, he has been leading several industrial and government projects. Dr. Zhang is a member of the China Power Electronics Society.